

Business performance measurement – past, present and future

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Abstract

Business performance measurement (BPM) is a fast evolving and diverse research field which features highly on the agenda of academics and practitioners from functions including general management, accounting, operations research, marketing, and human resources. Utilizing a citation analysis this paper identifies the following challenges for the field of BPM. The balanced scorecard seems to be the most influential and dominant concept in the field. Researchers are encouraged to further test and discuss its theoretical foundation and research methodology. The second challenge is to create a cohesive body of knowledge in the field of BPM.

Business performance measurement (BPM) is on the radar screen of business managers and academic scholars alike. Special issues of journals appear regularly and Harvard Business Press as well as Cambridge University Press have recently published collections of articles on measuring corporate performance. New reports and articles on the topic have been appearing at a rate of one every five hours of every working day since 1994 (Neely, 2002). Internet searches on the topic reveal more than 12 million sites dedicated to BPM. Furthermore, the software market for solutions and applications for measuring and managing corporate performance is constantly growing (Marr and Neely, 2001).

Like in many emerging research areas developments are rapid. Recent years have seen the development of new approaches of measuring performance, such as activity-based costing (Kaplan and Cooper, 1997) and shareholder value (Rappaport, 1986). New measurement frameworks, most notably the balanced scorecard (BSC) (Kaplan and Norton, 1992, 1996a) and assessment frameworks such as the business excellence model, have taken the business community in storm. Research suggests that 60 per cent of *Fortune* 1000 companies have experimented with the BSC (Silk, 1998). Other frameworks include the performance pyramid (Lynch and Cross, 1990), the macro process model (Brown, 1996) and more recently the performance prism (Neely *et al.*, 2002). Moreover, the recognition of non-financial and intangible assets has led to the development of various frameworks which address this evermore important area (Roos *et al.*, 1997; Edvinsson and Sullivan, 1996; Stewart, 1997; Sveiby, 1997; Brooking, 1996; Marr and Schiuma, 2001; Lev, 2001).

Academic research in the field of BPM comes from a wide variety of different

functions and disciplines, including accounting, economics, human resource management, marketing, operations management, psychology, and sociology. The growing level of interest in this field of research begs the question whether BPM should be a research discipline in its own right. In this paper we explore this field and take a look at the past, present, and future of research in the field of BPM. The aim of this paper is to empirically investigate and reflect upon BPM as a research field by utilizing a citation analysis to identify which scholars predominantly influence the field, understand which publications and concepts underpin the field, understand which management disciplines contribute the field. This analysis is then used to identify some challenges for future research in this fast evolving area of research.

Methodology for the citation analysis

For this study we conducted a citation analysis of the BPM field using 301 papers published in the PMA conference proceedings between 1998 and 2002 with a total of 4,464 citations. The 1998 proceedings *Performance Measurement – Theory and Practice* included 92 papers with a total of 1,424 references. The 2000 proceedings *Performance Measurement – Past, Present and Future* contained 94 papers with a total of 1,279 references. And the 2002 proceedings *Performance Measurement – Research and Action* included 115 papers with a total of 1,761 citations.

The multidisciplinary nature of the field of BPM made it difficult to select a group of journals appropriate for this analysis (Neely, 2002). Research papers of the PMA conferences represent a good research



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opportunity as the conferences were designed to integrate this multidisciplinary field. The papers analyzed include the research on performance measurement from over 400 authors in about 30 different countries and from over 200 institutions. The double blind review process with a paper rejection rate of over 30 per cent ensured rigor and quality of the research papers included in the analysis. However, the limitations of such an analysis are that we are unable to control of any bias in terms of researcher community who attended the conference. Not all disciplines are equally represented at conferences and especially at the first and second PMA conference there seemed to be a slight bias towards the operations management community. Furthermore, all conferences had a bias towards European researchers.

Utilizing a citation analysis the research set out to understand which authors and which conceptual ideas form the major foundations of the field of BPM. The method of citation analysis consists of counting and ranking the number of times documents are cited in bibliographies (Baker and Lancaster, 1991). Citation analysis is a recognized methodology to examine the body of literature influencing a discipline or field of academic research (Culnan, 1986). The underlying assumption is that more frequently cited articles have a greater influence on the academic field in comparison to articles cited less frequently (Todorov and Glanzel, 1988; Sharplin and Mabry, 1985).

Citation analysis has been a tool of research in the psychological literature for many years but also has been subject to criticisms for a number of reasons. For example, authors may not cite all the works that influenced the development of concepts, citations may be made to carry favor with important people, increase the length and "scholarliness" of papers, or to increase the number of self-citations or citations to friends (Sylvia, 1998). Furthermore, it is sometimes noted that these studies do not distinguish between positive and negative evaluations of cited references (Cohn and Farrington, 1990). However, Cole (1975) shows in a case study that the vast majority of citations were either positive or neutral, only 6 per cent of citations were negative. There seems to be consensus among citation researchers throughout the social sciences that authors and works are seldom cited for the purposes of criticism (Chapman, 1989; Garfield, 1979). Another deficiency, as noted by White (1984) is the unreliability of data

sources and inaccuracies in citations (Boyce and Banning, 1979).

Self-citations cause the following problems. Excluding self-citations increases the risks of underestimating the influence of prolific authors, responsible for numerous publications in a field. Including self-citations on the other hand bears the risk of overestimating the influence of writers who are fond of citing their own work (Wright and Miller, 1998). The citation analysis for this research has been carried out excluding all self-citations as analysis results confirmed that there was no risk of underestimating influential authors with the existing data set.

The well-defined concerns surrounding citation analysis discussed above and as raised in Pilkington and Listons-Heyes (1999) are acknowledged but adequate screening and sufficiently large sample size of our study ensure the validity of this citation analysis. Overall a citation analysis represents a useful means to understand the foundations of ongoing research in a specific scientific field (Vokurka, 1996). It provides guidance to what theories, authors and papers, can be considered as having a major impact on the field (Todorov and Glanzel, 1988). It therefore allows us to gain an understanding of which authors and papers have made a significant impact along the evolution of a field of research, in this case the field of BPM.

Analysis results

In the following section of this paper we summarize the findings of the analysis in terms of cited authors, cited references, as well as cited journals.

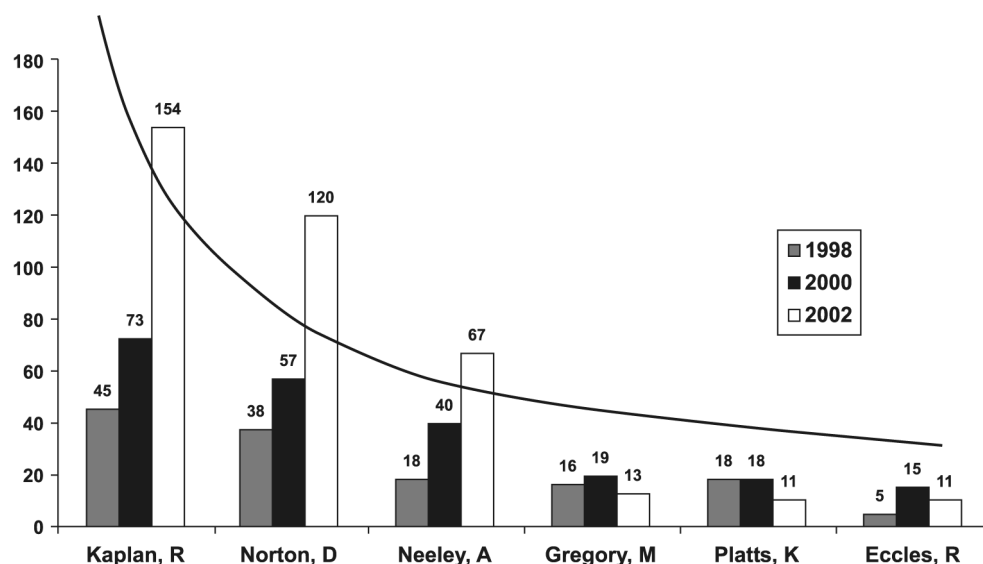
Cited authors

The citation analysis shows that there are few authors which are frequently cited and the majority of authors are cited only once or twice. The distribution is strongly skewed and the field seems to be very diverse with over 95 per cent of all authors referenced only once or twice.

In all three years Robert Kaplan is the most cited author with 45 citations in 1998, 73 citations in 2000, and 154 citations in 2002 (see Figure 1). David Norton and Andy Neely[1] are the second and third-most cited authors. In fact over 70 per cent of all papers quoted Kaplan and Norton in 2002, which increased from 45 per cent in 2000 and 34 per cent in 1998; 40 per cent of all papers cite Andy Neely in 2002, whereas in 2000 38 per cent and in 1998 15 per cent did so. The number of

Figure 1

Most frequently cited authors by year



citations for Kaplan, Norton, and Neely have steadily increased. Other frequently cited authors (Mike Gregory, Ken Platts, Robert Eccles) were cited proportionately less in 2002 than in 2000 and 1998.

Cited references

The strongly skewed distribution is repeated for referenced papers. Tables I-III show all papers cited more than one or two times for each year. Few papers seem to significantly impact the field whereas most other papers are cited only once. In fact, only 6 per cent of all citations are referenced more than once and in 2002 only 26 references are cited more than two times (20 in 2000 and 18 in 1998). This leaves a total of 1,177 articles that are referenced only once in 2002 alone.

In all three years Kaplan and Norton's 1996 book *The Balanced Scorecard – Translating Strategy into Action* as well as their 1992

Harvard Business Review article introducing the BSC are the two most cited references. The third most cited work in 2000 is the 1998 book *Performance Measurement – Why, What and How* by Andy Neely. In 1998 the 1995 joint paper "Performance measurement system design: a literature review and research agenda" by Neely *et al.* comes third sharing third place with the 1996 Kaplan and Norton *Harvard Business Review* paper "Using the balanced scorecard as a strategic management system". In 2002 Kaplan and Norton dominate the first five places with only the fifth place shared with Neely and Adams' (2001) article "The performance prism perspective".

The analysis implies that the concept of the BSC has impacted the field most significantly. The frequency of papers citing Kaplan and Norton or the BSC has increased over the last six years. In 2002 almost three-quarters of all papers refer to the BSC

Table I

Most frequently cited references in 2002

2000 papers	Citation – frequency
Kaplan and Norton (1996a)	35
Kaplan and Norton (1992)	26
Kaplan and Norton (2000b)	19
Kaplan and Norton (1996c)	12
Neely and Adams (2001), Kaplan and Norton (1996b, 1993)	8
Olive <i>et al.</i> (1999), Eccles (1991)	7
Sveiby (1997), Johnson and Kaplan (1987)	6
Neely <i>et al.</i> (1995), Kennerley and Neely (2000), Kaplan and Norton (2000a), Lynch and Cross (1991), Lev (2001), Keegan <i>et al.</i> (1989)	5
Neely (1998; 1999), Roos <i>et al.</i> (1997), Brooking (1996), Stewart (1997), Maskell (1991), Ittner and Larcker (1998), Yin and Campbell (1994), Prahalad and Hamel (1990)	4

Table II
Most frequently cited references in 2000

2000 papers	Citation – frequency
Kaplan and Norton (1996a)	23
Kaplan and Norton (1992)	18
Neely (1998)	10
Kaplan and Norton (1996c)	9
Keegan <i>et al.</i> (1989), Neely <i>et al.</i> (1997), Roos <i>et al.</i> (1997), Kaplan and Norton (1996b), Neely <i>et al.</i> (1995), Kaplan and Norton (1993)	5
Yin and Campbell (1994), Eccles and Pyburn (1992), Dixon <i>et al.</i> (1990)	4
Nanni <i>et al.</i> (1992), Bititci <i>et al.</i> (1997), Deming (1986), Epstein and Manzoni (1997), Senge (1990), Cross and Lynch (1989), Sink (1991)	3

Table III
Most frequently cited references in 1998

1998 papers	Citation – frequency
Kaplan and Norton (1996a)	14
Kaplan and Norton (1992)	12
Neely (1998), Kaplan and Norton (1996c)	7
Dixon <i>et al.</i> (1990)	5
Emmanuel <i>et al.</i> (1990), Senge (1990), Eccles (1991)	4
Skinner (1985), Hayes and Wheelwright (1984), Hamel and Prahalad (1993), Narver and Slater (1990), Lynch and Cross (1990), Ghalayini <i>et al.</i> (1997), Yin and Campbell (1994), Porter (1985), Womack and Jones (1996), Beischel and Smith (1991)	3

concept and six out of the ten most cited papers are by Kaplan and Norton on the BSC. Whereas in 2000 half of all papers cite the BSC and in 1998 only three out of the ten most cited references are by Kaplan and Norton.

Cited journals

The most frequently cited journal is *Harvard Business Review* in all years, followed by the *International Journal of Operations & Production Management*, *Journal of Marketing*, *Journal of Management Accounting Research*, and *Strategic Management Journal* (see Figure 2). A total of 321 different journals were cited in 2002 which increased from 218 journals in 2000 and 272 journals in 1998.

Discussion and challenges

Taking into account the limitations of this study the analysis indicates that the field of BPM is very diverse with a broad range of authors, papers, and disciplines contributing to this academic field. At the same time the distribution off each analysis is heavily skewed which indicates that only a few authors and a few concepts heavily influence this cross-functional field. Many others make contribution, however with little impact to the field as a whole. Kaplan and Norton are the dominating authors in the field. The BSC

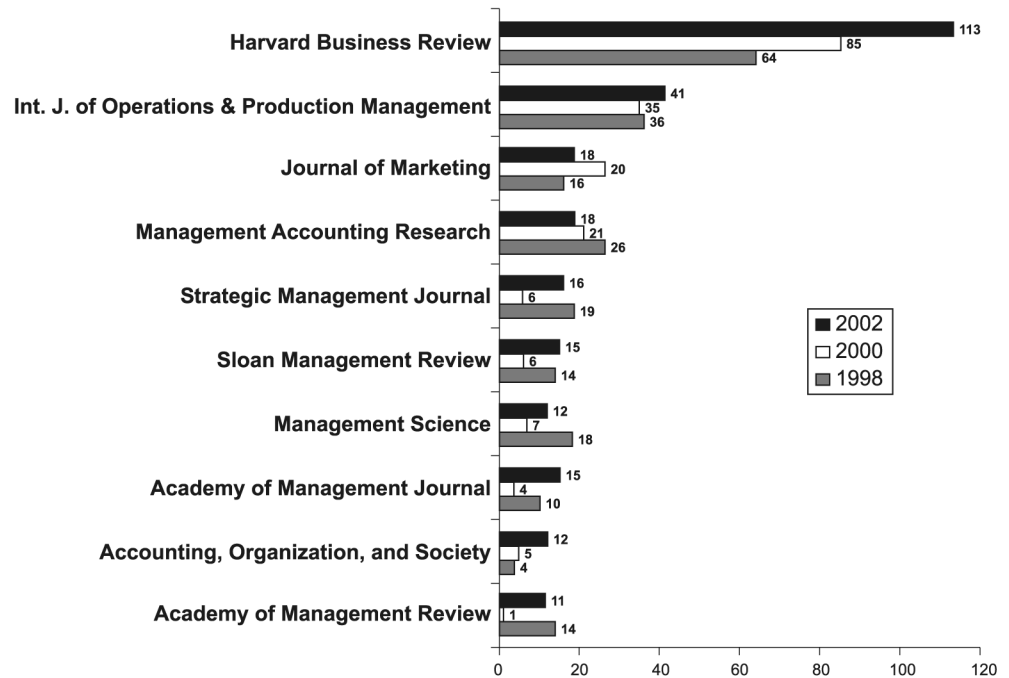
introduced by Kaplan and Norton is the dominating concept in the field of BPM followed by the performance prism concept and more general works on performance measurement by Neely *et al.*

The BSC lends itself to cross-functional citations as it brings together seemingly disparate elements of corporations such as finance and accounting, marketing, operations management, as well as HR and innovation research (Kaplan and Norton, 1992). Furthermore, the BSC seem to have constantly evolved from a measurement framework in 1992, to a management framework in 1996, and to an organization and change framework in 2000 making it even more appealing to a broader audience (Kaplan and Norton, 1992, 1996a, 2000b). Some scholars raise the question that there is little theoretical foundation of the BSC concept and others criticize the implied causal relationships, or the usability of the BSC (Lipe and Salterio, 2000; Norreklit, 2000; Brignall, 2002; Lipe and Salterio, 2002).

The BSC is based on “innovation action research” (Kaplan, 1998). Innovation action research is a methodology that is not yet widely discussed, used, or accepted but the BSC, which is grounded in this methodology, seems to be well accepted. A precondition of innovation action research is a major limitation in current practice and it is therefore hard to find and use organizations

Figure 2

Most frequently cited journals by year



to discover and validate theories (Kaplan, 1998). However, it would be interesting to discover how ideas such as the *Tableau de Bord* influenced the BSC development. The *Tableau de Bord*, for example, emerged in France as far back as the turn of last century and represents a management dashboard developed to better understand causal relationships between actions and processes (Epstein and Manzoni, 1997).

Much of the BSC research is case-based and it seems that there is a lack of large-scale empirical testing of the concept. The empirical study of 132 respondents conducted by Frigo and Krumwiede (1999), for example, does not prove the breakthrough success often reported in case study research. Here seems to be immense opportunities for further empirical tests of concepts that underlie the field of BPM, such as the BSC and strategy maps (Kaplan and Norton, 2000a). It might well be that large-scale empirical studies do not allow us to disentangle the context-specific nature of successful implementation in the same way rich case studies do, however, further discussion around research methodology would be useful.

The journal analysis showed that academic work in BPM is published in a diverse set of journals and indicates that it comes from a wide range of disciplines. Following the same logic and taking the journal topics as an indicator would imply that the main disciplines impacting the field of BPM are

strategic and general management, operations management, marketing, and finance and accounting. Other frequently cited journals come from disciplines such as economics, human resource management, organizational behavior, and public sector management. This diversity of disciplines raises the question where and whether there is a coherent body of knowledge being created for the field of BPM? Academic disciplines such as marketing or strategy have dedicated journals, which create such a body of knowledge in an on-going dialogue. This does not apply for cross-disciplinary fields such as BPM. Even though there are dedicated journals such as the *International Journal of Business Performance Management* or *Measuring Business Excellence* they do not, as yet, impact the field significantly. It seems that without a cohesive body of knowledge and dedicated high-level journals it is difficult to build a common theoretical foundation of the field of BPM.

Whereas diverse and multi-disciplinary of research is appealing it also carries the danger of hindering developments in the field of BPM (Neely, 2002). It makes it difficult for researchers to build upon a body of knowledge created by previous researchers because contributions are scattered around in literature across different disciplines. Academic disciplines often operate in functional silos but those go against the efforts to integrate knowledge in order to create a cohesive view and understanding of

organizations (Neely *et al.*, 2002). Marketing researchers could end up with their marketing measurement framework and HR researchers could develop an isolated HR scorecard. Therefore another challenge the field of BPM faces in the future is the creation of a body of knowledge that reflects the theoretical foundations of BPM as an autonomous research field.

Summary and conclusion

In summary there are two issues emerging from this citation analysis. The first is the dominance of the BSC concept. The BSC seems to be the most influential concept in this field but there appears to be a significant lack of information about its theoretical foundation. The field of BPM would therefore contribute from further discussions about the theoretical underpinnings and research methodologies of the BSC. In addition more empirical research aimed to validate the theories put forward in the BSC are encouraged. The second issue is the lack of a cohesive body of knowledge in the field of BPM. There seems to be scope for integrating literature in order to avoid replications of research in academic silos and isolated solutions. This citation analysis indicates that only few concepts and authors manage to bridge the disciplinary boundaries in order to influence the wider field of BPM. Active involvement in the PMA might facilitate the development of a more cohesive body of knowledge in order to develop the field of BPM into a research discipline of its own right.

Note

- 1 At this point we would like to acknowledge the deficiency that we were unable to control for the fact that Neely chaired the conferences in 1998 and 2000, and co-chaired the conference in 2002, which might have impacted the citations.

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